stabilis — progress report

Frequency stability analysis engine. Milestones M0, M1 and M2 complete — the v0.1 line, the first version from which an uncertainty may legitimately be quoted — plus three pieces of M3: Greenhall EDOF, which is error-bar parity with Stable32; the exact non-negative least squares core; and the power-law decomposition built on it.

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`commit 8ea48bb (20 total) — WORKING TREE DIRTY · every figure below is measured at generation time`

What this is

Stable32 is the de-facto tool for frequency stability analysis. It is Windows-only, closed, GUI-driven, unscriptable and no longer developed. *stabilis* is an open-core engine for the same job, intended as the free foundation under a paid workbench that produces signed ISO-17025-style reports for calibration labs and oscillator manufacturers.

Two things distinguish it from a reimplementation. First, the Haar wavelet variance *is* the Allan variance, so classical AVAR is the K=1 case of one framework; higher-order Daubechies filters then buy drift immunity and convergence for steeper noise. Second, confidence intervals are derived from the estimator's own coefficient autocovariance, so they do not require identifying the noise type first — Stable32 looks it up in a table and has no answer when the noise is not one of its five power laws.

> **The commercial argument for validation.** ISO/IEC 17025:2017 clause 6.4.1 defines software as *equipment*, and 7.11.2 requires laboratory software to be validated before introduction. Its NOTE 2 exempts commercial software "in general use within its designed application range" — an exemption stabilis fails twice over, being new and being open-core, since a lab that patches it explicitly triggers re-validation. An accredited lab therefore cannot adopt this engine unless it ships validation evidence. That evidence is the product's moat, not its overhead.

Results: accuracy against certified values

Every classical variance is a sum of squares of integer linear combinations of the phase samples, divided by an integer. Both NBS14 reference sets are integer-derived, so on them the variance is an **exact rational**. It is computed independently in `fractions.Fraction`, with the deviation certified to 50 digits via `decimal.Decimal.sqrt()`. That gives certified values computed here, to far more digits than any published table, with no dependence on a transcription being correct.

Accuracy is reported as **LRE**, the log relative error, after the NIST Statistical Reference Datasets convention: it is the number of correct significant digits. float64 carries about 15.95, so these are at the representation limit.

Estimator	NBS14 9-point (min LRE)	NBS14 1000-point (min LRE)
adev	16.3	14.8
oadev	16.1	14.8
mdev	16.3	14.1
tdev	16.0	14.1
ohdev	16.2	14.5

A worked certified value

Overlapping Hadamard variance of the NBS14 nine-point set at AF 1, as an exact rational and as a 50-digit deviation:

exact variance	`10027 / 2`
certified deviation	`70.806073185850378720054399356481405765799283518283`
computed (float64)	`70.80607318585038`

> **The oracle earned its keep immediately.** The relayed NBS14 table gave `hdev(1) = 70.80608` while listing `ohdev(1) = 70.80607` — but these are the same estimator at AF 1, since there is no overlap to exploit. The certified value settles it at `70.806073185850378`, so the first was a transcription error. This is exactly why correctness rests on algebra rather than on published tables.

Results: analytic identities

These need no reference data and no oracle, which is why they carry the correctness argument. Every published ADEV table traces back to Stable32, so agreeing with one proves interoperability; agreeing with algebra proves correctness.

Identity	Claim	Result
linear frequency drift: avar = $D^2 \cdot \tau^2/2$	bit-exact	**PASS**
linear frequency drift: hvar = 0	bit-exact	**PASS**
constant frequency offset leaves avar unchanged	bit-exact	**PASS**
scale equivariance at 2^k: var → 64·var for c = 8	bit-exact	**PASS**
adev = oadev = mdev at AF 1	rel 1e-15	**PASS**
tdev = $\tau/\sqrt{3}$ · mdev	rel 1e-15	**PASS**
constant phase gives exactly zero	bit-exact	**PASS**

> **The most valuable single test.** For a pure linear frequency drift D, the phase is quadratic, so the second difference is the constant $D \cdot \tau^2$ and the third difference vanishes. Hence `avar = D²τ²/2` and `hvar = 0`, both exactly. One test validates the Allan normalization, the Hadamard normalization and the drift convention simultaneously — and it is bit-exact, not approximate.

Results: interoperability with Stable32

Labelled as interop, not correctness: these published values were produced by Stable32 itself. Seven significant figures as published, so agreement is assessed at rtol 1e-6.

Estimator	AF	computed	published	rel. diff
adev	1	2.9223188e-01	2.9223190e-01	7.5e-08
adev	10	9.9657361e-02	9.9657360e-02	6.3e-09
adev	100	3.8978043e-02	3.8978040e-02	8.5e-08
oadev	1	2.9223188e-01	2.9223190e-01	7.5e-08
oadev	10	9.1599534e-02	9.1599530e-02	4.6e-08
oadev	100	3.2413430e-02	3.2413430e-02	8.0e-09
mdev	1	2.9223188e-01	2.9223190e-01	7.5e-08
mdev	10	6.1723764e-02	6.1723760e-02	6.2e-08
mdev	100	2.1709209e-02	2.1709210e-02	4.0e-08
tdev	1	1.6872015e-01	1.6872020e-01	2.8e-07
tdev	10	3.5636232e-01	3.5636230e-01	4.7e-08
tdev	100	1.2533818e+00	1.2533820e+00	1.8e-07
ohdev	1	2.9438833e-01	2.9438830e-01	9.9e-08
ohdev	10	9.5810832e-02	9.5810830e-02	1.8e-08
ohdev	100	3.2376383e-02	3.2376380e-02	7.8e-08

Results: reference generator and numerics

NBS14 1000-point set ships as arithmetic, not a file

The set is defined by a published Lehmer recursion `k[i+1] = 16807·k[i] mod 2147483647` from seed `1234567890`, so it needs no data file and carries no licence question. Verified against the published `wriley.com/tst_suit.dat`: all 1000 values agreed to 5.55e-16 — one ULP, which is that file's own text rounding. **The generator is the more exact artifact of the two.**

k[0]	`1234567890`
k[1] (check value)	`395529916` PASS
vs published digits	`max dev 3.33e-16`

Phase ↔ frequency round-trip

Recovers the offset-removed phase to a few ULP. Not bit-exact: a cumulative sum of differences does not telescope exactly in floating point. The test bounds this at 8 ULP, a figure derived from the measurements below rather than tuned.

n	peak (s)	max error (s)	ULP
10	2.690e-12	1.010e-28	0.2
1000	5.009e-11	6.462e-27	1.0
100000	2.415e-10	1.292e-25	2.5

Power-law noise synthesis

Kasdin–Walter generation, log-log periodogram slope recovered over 1e-3 < f < 0.1, averaged over eight realizations.

α (requested)	slope (recovered)	|error|
2	+1.9734	0.0266
1	+0.9912	0.0088
0	+0.0043	0.0043
-1	-0.9943	0.0057
-2	-1.9955	0.0045

Results: one missing sample

Gap handling is where a metrology tool fails most expensively, because it fails quietly. Integrating frequency to phase with a cumulative sum cannot step over an unknown increment, so a single `NaN` made every later phase sample `NaN` — and the estimator, dropping non-finite terms as designed, silently analysed only the data *before* the gap.

Measured at generation time over 400 realizations of 200 white-FM points with one sample removed at the midpoint, each compared against its own ungapped answer. *Old* integrates the gapped record in one cumulative sum, which is what the estimator used to do; *new* segments at the gap, integrates each run from its own zero and pools the coefficients. Bias is the mean signed error and is the figure that matters — on any single record either path can land closer by luck, but only one of them is centred on the right answer.

m	old bias	new bias	old RMS	new RMS	terms ungapped / old / new	terms lost
1	-0.04%	+0.01%	5.92%	0.57%	199 / 99 / 197	2 = 2m
2	-0.50%	+0.03%	6.78%	0.77%	197 / 97 / 193	4 = 2m
4	-0.86%	-0.10%	8.82%	1.52%	193 / 93 / 185	8 = 2m
8	-1.25%	-0.18%	12.25%	3.27%	185 / 85 / 169	16 = 2m
16	-3.00%	-0.29%	18.16%	6.92%	169 / 69 / 137	32 = 2m

> **One sample in two hundred, and the old path was biased low by percent — with no warning and a plausible-looking term count.** The bias is systematic and grows with τ, because the surviving terms were not a random subset of the record: they were the first half of it. Scatter falls by roughly an order of magnitude too, which is the direct consequence of keeping 197 terms instead of 99.
> The correct cost of a gap is the `2m` terms whose window spans it — `3m` for the Hadamard and modified families — and that is now exactly what it costs, asserted term by term in `tests/test_gaps.py`. Frequency input defaults to the segmenting policy and refuses the masking one; phase input keeps masking, which is correct when the timescale ran continuously across the gap. The choice, the number of runs and the longest run are all recorded in provenance.
>
> The residual few tenths of a percent in the "new bias" column is not a defect and does not go away: `E[σ̂] = σ·√(2/η)·Γ((η+1)/2)/Γ(η/2)`, so a deviation estimated from fewer terms is genuinely low on average. It is the same bias the ungapped estimate carries, slightly amplified. Correcting it, and reporting the error bar that goes with it, is what M2 went on to do — and the reason this defect had to be fixed first, since it feeds the term count that sets η.

Design decisions worth recording

Scale factors live outside the coefficients

Every estimator returns the coefficient array it averaged, so one confidence-interval implementation can serve all of them. The obvious form is `variance = mean(c²)`, which requires dividing by √2 or √6. That was implemented and then reverted: it makes every coefficient irrational and destroys exactness where the mathematics is exact — the drift identity above failed by one bit. The contract is now `variance = scale · mean(c²)`. EDOF depends only on the scale-free correlation structure, so nothing is lost.

float64 phase is a genuine hazard at this precision

At an accumulated phase of 10^3–10^6 s the float64 spacing is 10^{-13}–10^{-10} s, which silently quantizes away exactly the picoseconds sub-nanosecond work cares about. Phase is therefore stored as offsets from an explicit epoch, and construction warns when the spacing approaches the smallest resolved step. The warning is recorded in provenance, not merely printed, so it survives into a report.

The classical estimators are not routed through the wavelet filter bank

The Haar-equals-Allan identity is the intellectual core of the library, but making it the *implementation* would put boundary handling and shift-equivariance on the critical path for no user-visible gain. Written directly, the identity becomes an independent check of two separate code paths instead of a tautology.

No pickle support, by design

`pickle.load` on a file arriving from a customer's test rig is arbitrary code execution. `.npy` covers the same need safely.

Current state

test suite	`620 passed in 167.65s (0:02:47)` PASS
wall time	`174.0 s`
latest commit	`8ea48bb — Score the Stable32 reference session, and read the EDOF gap out of it`

Three structural properties are enforced by tests rather than convention: the core computes with scipy and matplotlib absent; every theory page must declare its validation status and name a covering test; and built-in readers and estimators register through the same decorator a third-party plugin would use.

Built

- **M0** — immutable phase/frequency containers with the precision guard, provenance with a step log, the plugin registry, tau selection, Kasdin–Walter synthesis, delimited-text reading, theory pages, and a CLI.
- **M1** — `adev`, `oadev`, `mdev`, `tdev`, `ohdev` computed directly from phase, gap-aware, plus the exact-rational oracle and the Tier 0 identity suite.
- **M2** — η_1 and η_3 degrees of freedom estimated from the estimator's own coefficient autocovariance, the pure-numpy incomplete gamma and inverse χ^2, χ^2 confidence intervals, lag-1 noise identification with an MDEV-slope cross-check, the deviation bias correction, and `stabilis validate` emitting a versioned evidence report that is capable of failing. This is the v0.1 line: before it, no uncertainty could be quoted from the library at all.
- **M3, first piece** — Greenhall EDOF, transcribed from the Greenhall & Riley PTTI 2003 preprint: all four cases, three coefficient tables and the generalized autocovariance, plus Satterthwaite pooling over the runs of a gapped record. It is what Stable32's `Alpha`, `Min Sigma` and `Max Sigma` columns are computed from, so it is what makes an old report and a new one diffable line for line. It is selected explicitly and never by `auto`: assuming a power law has to be a recorded act, and where the noise cannot be identified from the data the library declines and falls back rather than guessing, which is a place it deliberately answers less than Stable32 does.
- **M3, second piece** — the **power-law decomposition**, which reads a measured $\sigma(\tau)$ curve back into the five IEEE 1139 noise levels and a drift term. Because every h_a is a noise power and drift enters as D^2, the model is linear with a non-negativity constraint, which is exactly non-negative least squares — solved exactly by Lawson–Hanson, with a KKT optimality certificate on every fit, rather than by an optimizer that could be blamed for a negative answer. It fits Allan and *modified* Allan rows together, because white PM and flicker PM differ under AVAR only by a slowly varying logarithm: on noiseless data that is enough, but with 2% noise on the curve an AVAR-only fit returned between 0.00 and 1.00 times the true h_1 across six seeds, where the stacked fit held to 5% on every one. The conversion coefficients behind it were transcribed from NIST SP 1065 §7.1, whose flicker-PM row is printed two mutually inconsistent ways — resolved by integrating the handbook's own transfer function, and recorded rather than silently corrected. The five modified-Allan coefficients the handbook does not tabulate at all were derived here from its eq. (66).
- **Amplitude-calibrated synthesis** — the noise generator's h_α is now a settable parameter in SI units rather than an arbitrary scale, checked against four closed forms that are exact at every

averaging factor. That is what makes the decomposition testable end to end: noise is generated at a known level, estimated, decomposed, and compared to what went in.

Next

- **M3, in progress** — Greenhall EDOF, the NNLS core and the power-law decomposition have landed, and the **Stable32 reference session has begun** — real 1.62 exports over six synthetic records are committed and scored automatically, so the claim is no longer "implements the published algorithm" but "matches Stable32 to the precision it prints". Still to come: the MODWT wavelet layer, TOTVAR with its bias correction, and the rest of that session — the public NBS suites and, most valuable of all, a real oscillator record, being the only case where the data was not generated by the tool under test.
- **M5, first piece** — the **bench model**, which describes the actual experiment as sources, devices and channels, and turns three-cornered hat, N-cornered hat and instrument noise-floor subtraction into one non-negative linear solve per τ rather than three features. The part with no equivalent anywhere is the refusal: identifiability is a graph property — a connected set of measurements determines its oscillators individually *iff it contains an odd cycle*, which is why the technique is called *three*-cornered hat — so a ring of four determines nothing however long it runs, and `stabilis bench check` says so *before any data is collected* and names the one extra measurement that would fix it. Negative variances are reported signed rather than clipped, because they are the method telling you an independence assumption failed.
- **M4–M8** — model fitting against the χ^2 likelihood and instrument readers; then the rest of M5, cross-variances and multi-dataset handling; then realtime ingest with attributed time tagging and dynamic Allan variance; then the remaining parity tail, mask testing and the report layer.

> **Known limitations.** Confidence intervals now exist, but their measured coverage falls short of nominal at the extremes of the τ range; the measured figures live in `docs/validation-report.html`, generated by `stabilis validate`, which is the artifact that carries them. Greenhall's bounds are now available alongside the data-driven ones, but have now been checked against Stable32 1.62's own exports on this project's six synthetic records: the term-count column agrees exactly as integers, the sigma columns of all five estimators agree to 4.9e-5 — the last digit Stable32 prints — and Min/Max Sigma agree to 5.5e-4. The degrees of freedom behind those bounds do *not* quite agree: Stable32's implied EDOF runs about 1.2% above ours for averaging factors up to 32, where an independent implementation of the same paper returns our value to every printed digit. Cause unknown, worth under 0.06% on a 68.27% bound, and recorded rather than guessed at. No real oscillator record has been through both programs yet, which is the remaining part of that item in `VALIDATION-DEBT.md`. Of the ten power-law conversion coefficients, nine are checked end to end against noise of known amplitude; the tenth, unmodified flicker PM, cannot be with the current generator, because that coefficient assumes a brick-wall bandwidth and the Kasdin–Walter process is not brick-wall limited — the gap is measured, bounded and recorded rather than averaged away. Instrument readers are not yet validated against real hardware exports. TOTVAR, Theo1, MTIE, the wavelet layer and the bench model are not implemented.